NUMBER RACK (REKENREK) ARITHMETIC STUDIO

Dual-Row 20-Bead 5-and-10 Grouping & Subitizing Frame

SERIES 1 • CARD 11 / 12 • DOMAIN: 5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

CARD 11 / 12 • RACK-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

Counting by ones is slow and error-prone.
The Rekenrek organizes 20 beads into 5-anchors:
Row 1: 5 Red + 5 White • Row 2: 5 Red + 5 White
Sliding beads in groups turns math into subitizing!

CALIBRATED 102.0 MM SLIDER DATUM

- Beads: 10 Red + 10 White = 20 Beads (14.0 mm ⌀)
- Slider Rods: 2x 102.0 mm Octagonal Rods
- Upright Towers: 2x 49.0 mm Snap-Fit Towers (7U)

5-AND-10 REKENREK: 7 + 6 = 13



DOUBLES-PLUS-ONE

- Red: 5 + 5 = 10 (Instant T
- White: 2 + 1 = 3 Units

Sum = 10 + 3 = 13

Zero finger counting!

2. KIT PARTS INVENTORY (27 PIECES)

COMPONENT	QTY	SPEC	COLOR
Red Beads	10x	14.0 mm Bead	Crimson Red
White Beads	10x	14.0 mm Bead	Classic White
Slider Rods	2x	102.0 mm Rod	Matte Charcoal
Upright Towers	2x	49.0 mm	Matte Charcoal
Storage Tray	1x	Nesting Tray	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal
Stage Lid	1x	Slider Lid	Matte Charcoal

TOTAL: 20 Beads + 4 Frame Parts + Tray + Storage Bin + Lid = 27 Parts

DUAL-ROW 5-COLOR STRUCTURE

- 5 Red + 5 White per row: Instant 5-anchor visual subitizing.
- Both Rows Active: Models doubles, near-doubles, and bridging.
- Zero-Rattle Carry: Beads slide into nested tray slots.
- High Contrast: Crisp color separation across rows.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Slide 10 red beads left and 10 white beads right on both rods. Place rods and towers into the nesting tray. Missing beads leave a visible gap. Snap lid shut.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Dual 5-and-10 grouping bridges perceptual subitizing directly into formal arithmetic operations.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: DUAL-ROW BRIDGING & DOUBLES PLUS ONE

Step-by-Step Hands-On Tutorial: 7 + 6 = 13 via 5-and-5 Doubles

SERIES 1 • CARD 11 OF 12 • DOMAIN: 5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

CARD 11 / 12 • RACK-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/number-rack/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 1.OA.C.6 (Addition Fluency)
- CCSS 2.OA.B.2 (Mental Math)
- TEKS 111.3.b.3.C (Compose/Deco...)

Grades K–3 Subitizing & Addition

GUIDING INQUIRY

When you slide 7 beads on top and 6 on bottom, how many red beads are on the left side altogether?

1 STEP 1: SLIDE 7 BEADS ON TOP ROW

Push all beads to the right side of the rack (resting zero).
Slide 7 beads to the left on the top rod in one push:
All 5 red beads + 2 white beads = 7 beads!
Notice: 5-anchor lets you see 7 instantly

2 STEP 2: SLIDE 6 BEADS ON BOTTOM ROW

Slide 6 beads to the left on the bottom rod in one push:
All 5 red beads + 1 white bead = 6 beads!
We want to find 7 + 6 without counting one-by-one.

3-4 STEPS 3 & 4: GROUP DOUBLES & REGROUP

3. GROUP RED BEADS: Look at the red beads on the left: 5 + 5 = 10 (Instant Ten)!
4. ADD WHITE BEADS: Look at the white beads: 2 + 1 = 3 units.
Total: 10 + 3 = 13! Therefore: 7 + 6 = 13!
Calculator Verification: Press Number Rack Subitizing Icon ([▦]) and run [7] [ENTER] [6] [+] → 13.0000.

5 CORE PRINCIPLE: DOUBLES MAKE MENTAL MATH EFFORTLESS

Instead of counting 6 fingers past 7, the Rekenrek decomposes 7 + 6 into (5 + 2) + (5 + 1). The two groups of 5 red beads instantly make 10, leaving 3 white beads. Addition becomes instantaneous visual pattern recognition rather than tedious memorization!

★ EXTENSION INVESTIGATION: SOLVE: 8 + 7 USING 5-AND-5

Slide 8 beads on top (5 red + 3 white) and 7 on bottom (5 red + 2 white). Group the 10 red beads first! What is the sum? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

STACKCALC ARCHIVAL INSTRUMENT LABS
Interactive Mathematics Manipulative Ecosystem • Mountain View, CA • BPA-Free

PARTS & REFILL PORTAL
stackcalc.io/parts

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

RACK-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 27 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California